

Week 9: Bayesian regression

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1 Packages required

```
library(rstan)
library(tidyverse)
library(tidybayes)
library(janitor)
```

2 Reminder: what is a Gompertz model?

In 1825 Benjamin Gompertz, an actuary in London, proposed the following two-parameter model for mortality:

$$\mu(x) = \alpha e^{\beta x}$$

where $\mu(x)$ is the instantaneous mortality rate (hazard) at age x . This model assumes that mortality increases exponentially with age, which is a pretty good assumption for adult mortality (above age 40 or so, with the exception of older ages).

3 Data

For the following two examples we're going to use death and population counts by age and sex for Ontario, sourced from the Canadian Human Mortality Database project. We can read the data files in directly from the URLs:

```
dd <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Deaths_1x1.txt", skip = 1)
dp <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Population.txt", skip = 1)
```

These data files are in 'wide' format. For our purposes it's going to be easier to work with in 'long format'. So let's do that and also clean some other stuff up:

```

dd <- dd |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "deaths") |>
  mutate(deaths = as.numeric(deaths), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

dp <- dp |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "pop") |>
  mutate(pop = as.numeric(pop), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

d <- dd |>
  left_join(dp) |>
  mutate(mx = deaths/pop,
         log_mx = log(mx))

```

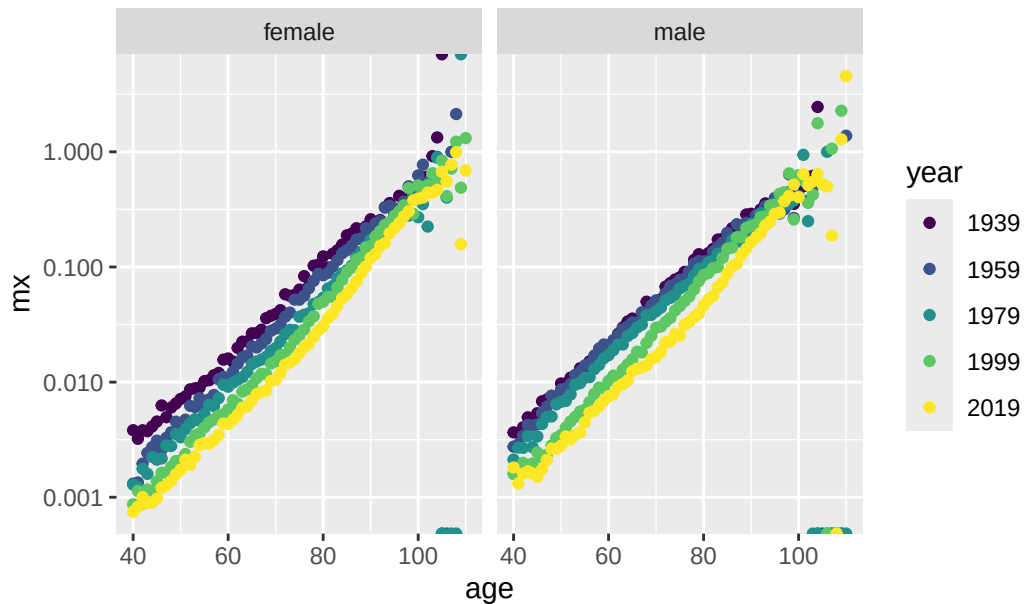
Do some quick plots

```

d |>
  filter(age>39, year %in% seq(1939, 2019, by = 20), sex!= "total") |>
  ggplot(aes(age, mx, color = factor(year))) +
  geom_point() +
  facet_wrap(~sex)+
  labs(title = "Mortality rate for Ontario by sex, 1939-2019")+
  scale_y_log10()+
  scale_color_viridis_d(name = "year")

```

Mortality rate for Ontario by sex, 1939–2019



4 Base model

First let's fit a Gompertz model to males aged 40+ in 2019. Note that Gompertz models have the form

$$\mu(x) = \alpha e^{\beta x}$$

So we can write

$$\log \mu(x) = \log \alpha + \beta x$$

More notes:

- To make this fully Bayesian we need to specify the likelihood and priors. The full model here is

$$y_x \sim \text{Poisson}(P_x \cdot m_x)$$

$$\log m_x = \log \alpha + \beta x$$

$$\log \alpha \sim N(0, 10^2)$$

$$\beta \sim N(0, 0.1^2)$$

where y_x is deaths at age x and P_x is population

- Could have used a normal likelihood (c.f. using `lm`) but nice to account for population size
- We are fitting not on age, but on a centered version (why?)

Now we need to get the data in the right format to read into Stan (this required a named list):

```
d_male_19 <- d |>
  filter(year==2019, sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

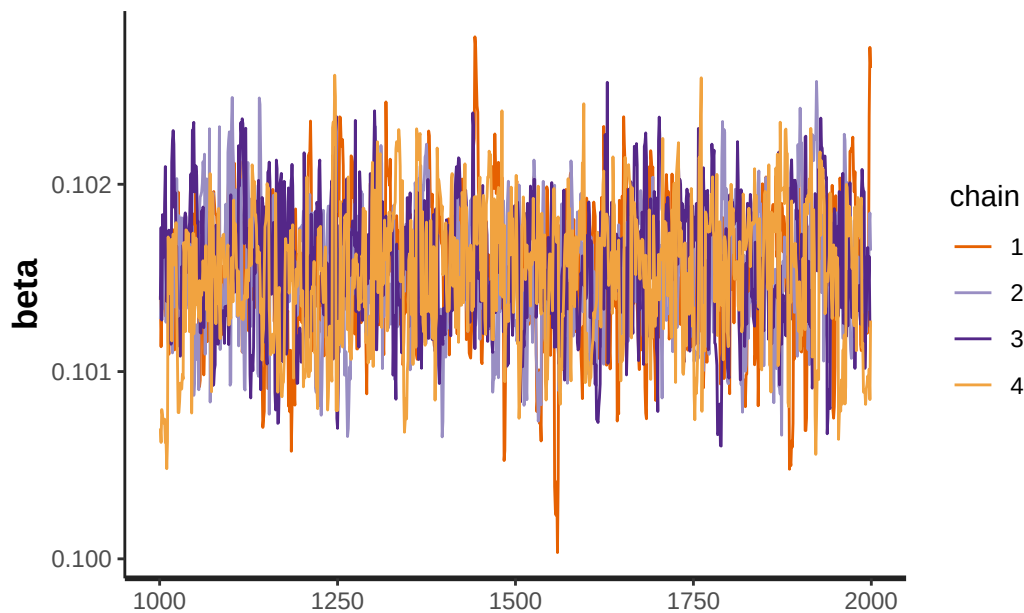
stan_data <- list(y = round(d_male_19$deaths),
                 pop = round(d_male_19$pop),
                 N = nrow(d_male_19),
                 age_c = d_male_19$age_c)
```

Run the model and look at some output:

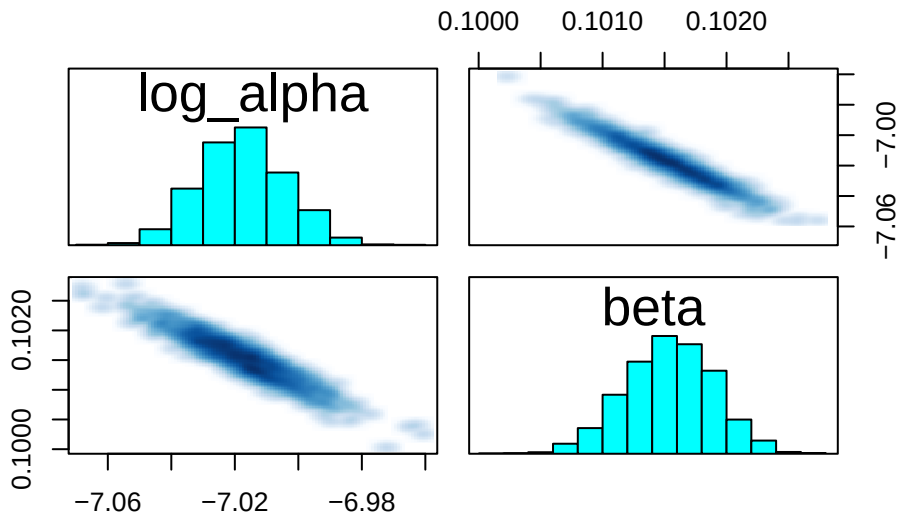
```
mod <- stan(file = "../code/models/gomp.stan",
            data = stan_data,
            seed = 123)
mod
```

Some quick model checks, looking at the traceplot and pairwise densities

```
pars <- c("log_alpha", "beta")
traceplot(mod, pars = c("beta"))
```



```
pairs(mod, pars = pars)
```



```
temp <- rstan::extract(mod)
names(temp)
```

```
[1] "log_alpha" "beta"      "mu"        "lp_"
```

```
temp$log_alpha
```

```
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 [3606] -6.989524 -7.019056 -7.024447 -7.039795 -7.036969 -7.027117 -7.016895
 [3613] -7.022166 -7.030846 -7.008295 -6.998507 -7.040560 -7.037164 -7.035834
 [3620] -7.010187 -7.013454 -6.995710 -7.027599 -7.031221 -7.014141 -7.011872
 [3627] -7.014464 -7.003115 -7.012084 -7.035170 -7.003850 -7.002849 -7.023425
 [3634] -7.009735 -7.026087 -7.038948 -7.027884 -7.000956 -7.016767 -7.032827
 [3641] -7.031545 -6.995926 -7.004771 -7.018488 -7.025223 -7.015572 -6.996924
 [3648] -6.989694 -7.014696 -7.000400 -7.047086 -7.028836 -7.014876 -7.010569
 [3655] -7.031509 -6.995605 -7.004506 -6.997530 -7.018718 -7.016567 -6.993147
 [3662] -7.044805 -6.998697 -7.017582 -6.987082 -6.997913 -7.018318 -7.027782
 [3669] -7.026414 -7.012952 -7.005592 -7.018363 -6.989569 -7.036403 -6.991136
 [3676] -7.034042 -7.024034 -7.040389 -6.991369 -7.012658 -7.016876 -7.006251
 [3683] -7.016799 -7.032934 -7.022952 -6.984303 -7.025863 -7.000681 -7.003997
 [3690] -7.020104 -7.011226 -7.028479 -7.034571 -7.030160 -7.003202 -7.007192
 [3697] -7.010094 -7.011060 -7.020973 -7.015761 -7.022449 -7.029862 -7.027818
 [3704] -7.034330 -6.997338 -7.018309 -7.015970 -7.000653 -7.031246 -7.025284
 [3711] -7.011825 -6.992290 -7.022865 -7.013525 -7.004251 -7.011868 -7.003159
 [3718] -7.021494 -7.011682 -7.017274 -7.026129 -7.010223 -7.026937 -7.030216

[3725] -7.006252 -7.019050 -6.996995 -7.003827 -7.021597 -7.027648 -7.024255
 [3732] -7.050389 -7.036969 -7.018764 -7.000046 -7.006047 -7.030190 -7.018452
 [3739] -7.003460 -7.007073 -7.024714 -7.006681 -7.011989 -7.008461 -7.000881
 [3746] -7.044215 -6.998513 -6.995347 -7.027818 -7.030907 -7.035075 -7.002819
 [3753] -7.019396 -7.022157 -7.019182 -7.025971 -7.022821 -7.020719 -7.029127
 [3760] -7.022093 -7.035931 -7.016087 -7.009936 -7.001849 -7.003577 -7.013241
 [3767] -7.020482 -6.988775 -7.042923 -7.009569 -7.006400 -7.013323 -7.016971
 [3774] -7.005998 -7.030397 -7.006241 -7.029477 -6.995056 -7.031688 -7.041153
 [3781] -6.987590 -7.024542 -7.004699 -7.023908 -7.031059 -7.003006 -7.004469
 [3788] -7.020060 -7.027654 -7.007702 -7.022502 -7.017156 -7.004986 -7.031347
 [3795] -7.031154 -7.008866 -6.998243 -7.033771 -7.013328 -7.021844 -7.030391
 [3802] -7.018543 -7.023348 -7.027488 -7.019397 -7.012346 -7.020800 -7.026515
 [3809] -7.042420 -7.010037 -7.008156 -7.024466 -6.991137 -6.998978 -7.010822
 [3816] -6.989922 -7.002240 -7.043165 -7.014604 -7.025384 -7.021614 -7.049616
 [3823] -6.999294 -7.012419 -7.049318 -7.020104 -7.016162 -7.019489 -7.014531
 [3830] -7.012604 -7.016004 -7.030320 -7.034279 -7.011070 -7.015065 -6.994516
 [3837] -7.031762 -7.029772 -7.027689 -7.005179 -7.023253 -7.009388 -7.018650
 [3844] -7.018326 -6.997563 -7.019377 -7.015731 -6.998977 -7.032695 -7.020973
 [3851] -6.990994 -7.022866 -7.036675 -6.998591 -7.018516 -6.989495 -7.022662
 [3858] -6.995784 -7.016067 -7.005003 -7.038374 -7.009079 -7.010485 -7.010953
 [3865] -7.011002 -7.012626 -7.005658 -6.996383 -7.025828 -7.027629 -7.030055
 [3872] -7.052399 -7.027014 -6.998764 -7.004884 -6.980852 -7.001526 -7.041180
 [3879] -7.026166 -6.989585 -7.046513 -7.016342 -7.019525 -7.036602 -7.008370
 [3886] -7.020917 -7.004085 -7.026007 -6.997585 -7.018678 -7.030159 -7.018691
 [3893] -6.990234 -7.026056 -7.020394 -7.012618 -7.018388 -7.034757 -7.016929
 [3900] -7.022531 -7.002490 -6.993899 -7.035063 -7.014408 -7.041919 -7.029584
 [3907] -7.018237 -7.009779 -7.008470 -7.013160 -7.017214 -7.010325 -7.017823
 [3914] -7.014269 -7.012401 -7.037443 -7.029011 -7.021684 -6.996968 -7.002472
 [3921] -6.996485 -7.003674 -7.010153 -6.991625 -7.020220 -6.994625 -7.014861
 [3928] -6.995263 -7.024432 -6.998684 -7.035112 -7.046246 -7.009115 -7.019817
 [3935] -7.021361 -7.032785 -7.013549 -7.004040 -7.021955 -7.018568 -7.027715
 [3942] -7.002926 -7.014418 -7.008499 -6.999717 -6.989385 -7.021450 -7.019775
 [3949] -7.022618 -7.024916 -7.025328 -7.017774 -7.014848 -7.012380 -6.986714
 [3956] -7.017543 -7.007709 -7.006164 -6.990185 -7.014007 -7.011141 -7.027513
 [3963] -7.009073 -6.998237 -7.027005 -7.007544 -7.019536 -7.025222 -7.011920
 [3970] -7.003561 -7.017438 -7.045216 -7.013149 -7.033051 -7.024931 -7.008612
 [3977] -7.017829 -7.027421 -7.036459 -7.010065 -7.014276 -7.045986 -7.021843
 [3984] -7.020884 -7.006203 -7.023236 -7.018194 -7.014328 -7.004131 -7.009710
 [3991] -7.025740 -7.038469 -7.021447 -7.000958 -7.031928 -7.022974 -6.999629
 [3998] -7.014627 -7.001467 -7.008736

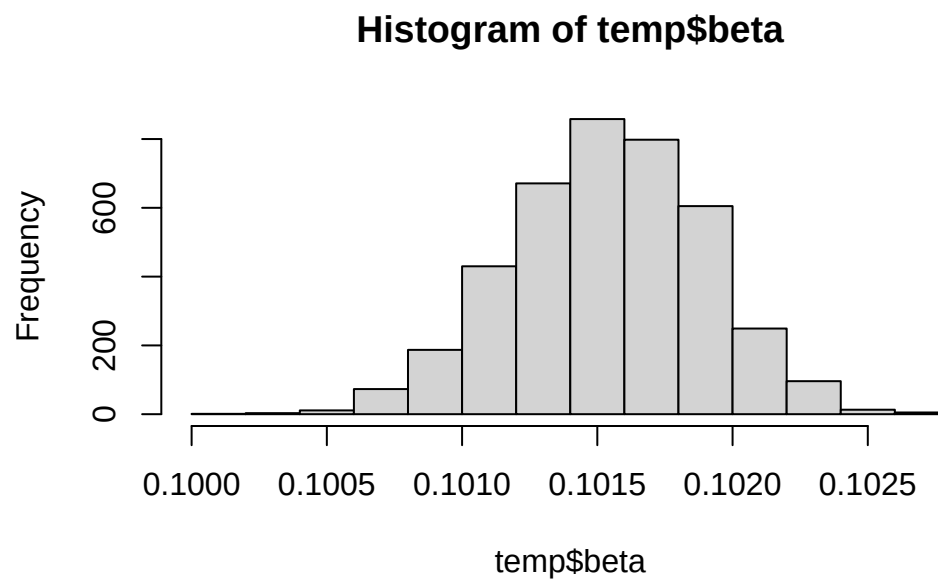
```
median(temp$log_alpha)
```

```
[1] -7.017539
```

```
median(temp$beta)
```

```
[1] 0.1015472
```

```
hist(temp$beta)
```



```
# 95% credible interval for beta  
quantile(temp$beta, 0.025) # lower
```

```
2.5%  
0.1008208
```

```
quantile(temp$beta, 0.975) # upper
```

```
97.5%  
0.1022141
```

4.1 Extract the parameter estimates using tidyverse

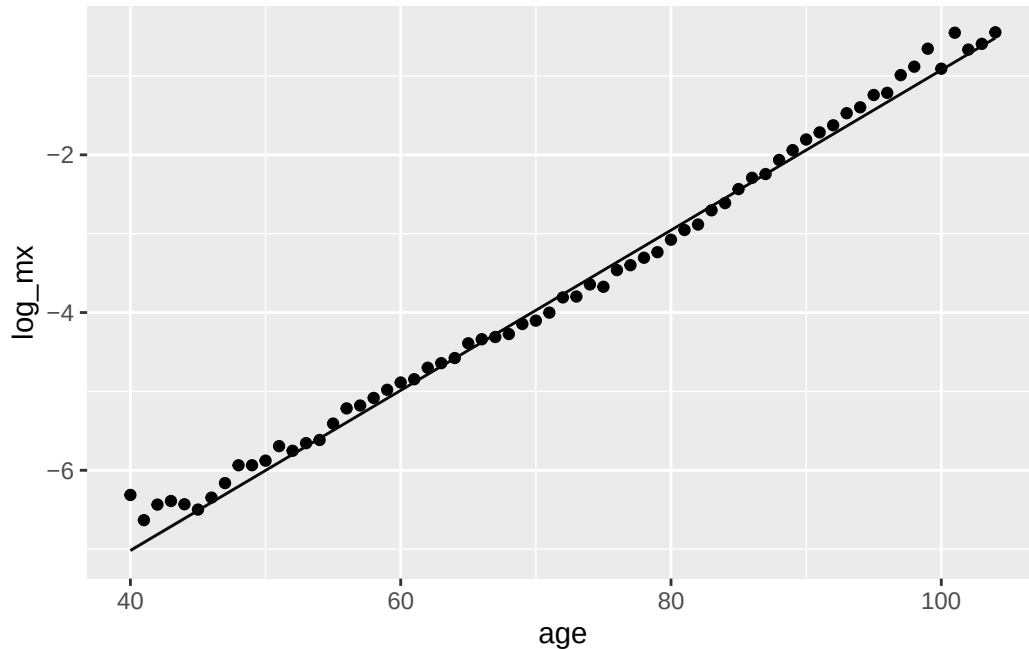
```
mod |>
  gather_draws(log_alpha, beta) |>
  median_qi()

# A tibble: 2 x 7
  .variable .value .lower .upper .width .point .interval
  <chr>      <dbl> <dbl> <dbl> <dbl> <chr> <chr>
1 beta      0.102  0.101  0.102  0.95 median qi
2 log_alpha -7.02  -7.04  -6.99  0.95 median qi
```

4.2 Calculate and plot some results

Now we can combine the `tidybayes` syntax with `ggplot` to plot some results. For example, here's the data versus the estimates for the linear predictor:

```
mod |>
  gather_draws(mu[x]) |>
  median_qi() |>
  mutate(age_c = x - 1) |>
  left_join(d_male_19) |>
  ggplot(aes(age, log_mx)) +
  geom_point() +
  geom_line(aes(age, .value)) +
  geom_ribbon(aes(x = age, ymin = .lower, ymax = .upper), alpha = 0.2)
```



Let's calculate the modal age at death (which for Gompertz mortality is a function of α and β , see [here](#)).

```
mod |>
  spread_draws(log_alpha, beta) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))) |>
  median_qi()
```

	log_alpha	log_alpha.lower	log_alpha.upper	beta	beta.lower	beta.upper
1	-7.017539	-7.043411	-6.990677	0.1015472	0.1008208	0.1022141

	mode_age	mode_age.lower	mode_age.upper	.width	.point	.interval
1	46.58586	46.49998	46.6706	0.95	median	qi

4.3 Question

Check you get similar results using `lm` or `glm`.

5 Model over time

Let's fit a slightly more complicated model, for multiple years, where the coefficients themselves are modeled as a random walk over time, i.e.

$$\beta_t \sim N(\beta_{t-1}, \sigma_\beta^2)$$

That is, a different set of Gompertz parameters are fit to every year, but we are assuming that the values in the current year are related to those in the previous year. This is a form of dynamic linear regression. We need to put priors on the first time point, and also on the variance terms:

$$\log \alpha_1 \sim N(-6, 1)$$

$$\beta_1 \sim N(0.1, 0.1^2)$$

$$\sigma^\alpha, \sigma_\beta \sim N^+(0, 1)$$

Note that the likelihood and model on mortality rates are as before, we just have an additional subscript for time:

$$y_{x,t} \sim \text{Poisson}(P_{x,t} \cdot m_{x,t})$$

$$\log m_{x,t} = \log \alpha_t + \beta_t x$$

Now to fit the model. First get the data in the right format:

```
years <- 1989:2019
d_male <- d |> filter(year>=years[1], sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

y <- d_male |>
  select(age, year, deaths) |>
  pivot_wider(names_from = "year", values_from = "deaths") |>
  select(-age) |>
  as.matrix()

pop <- d_male |>
  select(age, year, pop) |>
  pivot_wider(names_from = "year", values_from = "pop") |>
  select(-age) |>
  as.matrix()

stan_data <- list(y = y,
  pop = pop,
  N = nrow(d_male_19),
  age_c = d_male_19$age_c,
  T = ncol(y))
```

Now fit the model (note: takes a while):

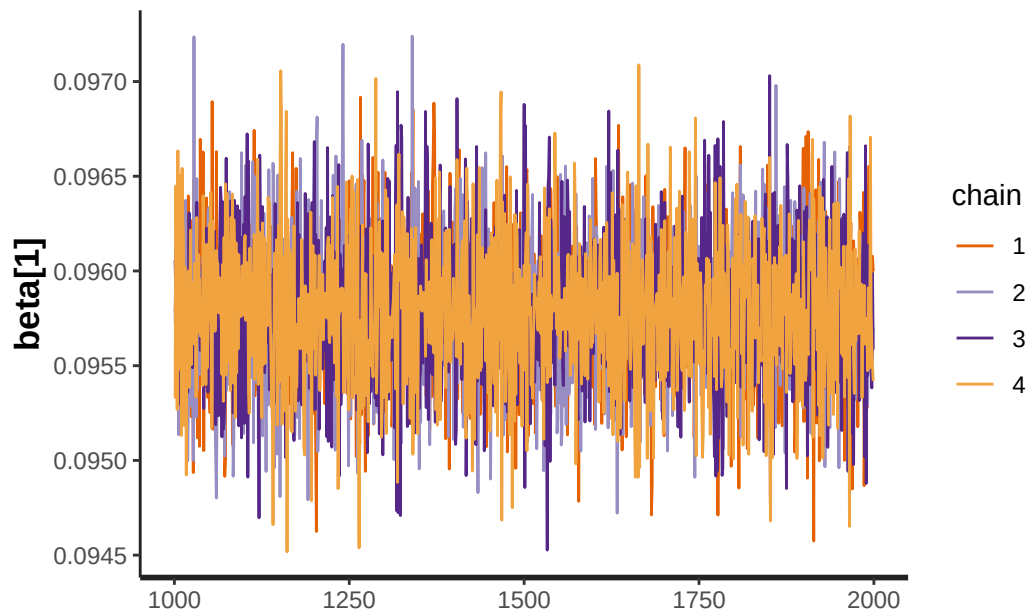
```
mod <- stan(file = "../code/models/gomp_time.stan",
  data = stan_data,
  seed = 852,
  refresh = 0)
```

```
summary(mod)$summary[paste0("beta[", 1:(length(years)), "]"),]
```

	mean	se_mean	sd	2.5%	25%	50%
beta[1]	0.09578793	4.944786e-06	0.0003844579	0.09504345	0.09552444	0.09578874
beta[2]	0.09561733	4.766850e-06	0.0003498153	0.09491913	0.09537802	0.09561861
beta[3]	0.09666647	4.843584e-06	0.0003372869	0.09602207	0.09643337	0.09666156
beta[4]	0.09687503	4.751516e-06	0.0003405176	0.09621448	0.09664190	0.09687918
beta[5]	0.09707253	4.866022e-06	0.0003406497	0.09642179	0.09684506	0.09707471
beta[6]	0.09803272	4.636630e-06	0.0003324452	0.09737592	0.09780716	0.09803110
beta[7]	0.09873439	4.784696e-06	0.0003405607	0.09807854	0.09850350	0.09873376
beta[8]	0.09982416	4.767882e-06	0.0003296936	0.09917550	0.09960669	0.09982553
beta[9]	0.10069938	4.674714e-06	0.0003338526	0.10004556	0.10047682	0.10070122
beta[10]	0.10201361	4.572435e-06	0.0003366903	0.10136433	0.10178519	0.10201354
beta[11]	0.10279273	4.706582e-06	0.0003411270	0.10214121	0.10255579	0.10279379
beta[12]	0.10218275	4.607462e-06	0.0003284196	0.10155630	0.10195738	0.10218752
beta[13]	0.10153681	4.580443e-06	0.0003301109	0.10089899	0.10131387	0.10153994
beta[14]	0.10108257	4.211452e-06	0.0003302915	0.10043361	0.10086317	0.10108024
beta[15]	0.10134916	4.313313e-06	0.0003197920	0.10073176	0.10112862	0.10134418
beta[16]	0.10127386	4.329052e-06	0.0003247967	0.10063199	0.10105376	0.10127524
beta[17]	0.10170251	4.462948e-06	0.0003215972	0.10107058	0.10148860	0.10170215
beta[18]	0.10161479	4.414507e-06	0.0003258945	0.10098219	0.10139095	0.10161333
beta[19]	0.10157848	4.707276e-06	0.0003167623	0.10096531	0.10136669	0.10157652
beta[20]	0.10206824	4.589046e-06	0.0003198271	0.10144379	0.10185710	0.10206288
beta[21]	0.10203889	3.989949e-06	0.0003080872	0.10144517	0.10182639	0.10203426
beta[22]	0.10197777	4.175615e-06	0.0003071387	0.10139065	0.10177088	0.10197885
beta[23]	0.10201547	3.830124e-06	0.0003107001	0.10139683	0.10181212	0.10201299
beta[24]	0.10191715	4.205477e-06	0.0003131010	0.10129728	0.10170963	0.10192000
beta[25]	0.10225264	4.105617e-06	0.0003018138	0.10165090	0.10205838	0.10225555
beta[26]	0.10247798	3.863254e-06	0.0003062194	0.10187650	0.10227629	0.10247945
beta[27]	0.10239813	4.232017e-06	0.0003068404	0.10179581	0.10218980	0.10239609
beta[28]	0.10161719	3.853299e-06	0.0003080245	0.10101124	0.10140778	0.10161972
beta[29]	0.10211793	4.170638e-06	0.0002952473	0.10153990	0.10191703	0.10211859
beta[30]	0.10173912	3.710148e-06	0.0002945693	0.10117425	0.10154471	0.10173761
beta[31]	0.10119722	3.936794e-06	0.0002973721	0.10062973	0.10099465	0.10119777
	75%	97.5%	n_eff	Rhat		
beta[1]	0.09604007	0.09655404	6045.086	0.9991689		
beta[2]	0.09585704	0.09630549	5385.360	0.9995745		
beta[3]	0.09690191	0.09731765	4849.148	0.9994215		
beta[4]	0.09710466	0.09754943	5135.879	0.9995767		
beta[5]	0.09730260	0.09772985	4900.812	0.9999517		
beta[6]	0.09825391	0.09869525	5140.854	0.9997199		
beta[7]	0.09897132	0.09939977	5066.178	0.9994587		

beta[8]	0.10004264	0.10046997	4781.563	0.9993806
beta[9]	0.10092154	0.10135216	5100.344	0.9997378
beta[10]	0.10223754	0.10268618	5422.083	0.9995478
beta[11]	0.10301778	0.10347756	5253.164	0.9999085
beta[12]	0.10240096	0.10282520	5080.828	0.9997388
beta[13]	0.10176247	0.10218526	5194.033	0.9999188
beta[14]	0.10130921	0.10173659	6150.794	1.0006784
beta[15]	0.10156645	0.10197577	5496.843	1.0004044
beta[16]	0.10149954	0.10189512	5629.084	0.9998982
beta[17]	0.10191832	0.10233324	5192.554	0.9998424
beta[18]	0.10183527	0.10224081	5449.915	0.9997355
beta[19]	0.10178999	0.10221313	4528.221	0.9997925
beta[20]	0.10228002	0.10270831	4857.196	1.0000978
beta[21]	0.10224725	0.10264472	5962.283	0.9995414
beta[22]	0.10219011	0.10257526	5410.386	0.9996067
beta[23]	0.10222234	0.10262599	6580.473	0.9997548
beta[24]	0.10212567	0.10254180	5542.916	0.9994616
beta[25]	0.10244993	0.10284213	5404.074	1.0011652
beta[26]	0.10267760	0.10307621	6282.882	0.9998761
beta[27]	0.10260781	0.10299480	5256.905	0.9999068
beta[28]	0.10182975	0.10221271	6390.062	0.9998297
beta[29]	0.10231600	0.10269657	5011.490	1.0002333
beta[30]	0.10193408	0.10232632	6303.657	0.9994245
beta[31]	0.10139169	0.10178508	5705.780	0.9997203

```
traceplot(mod, pars = c("beta[1]"))
```

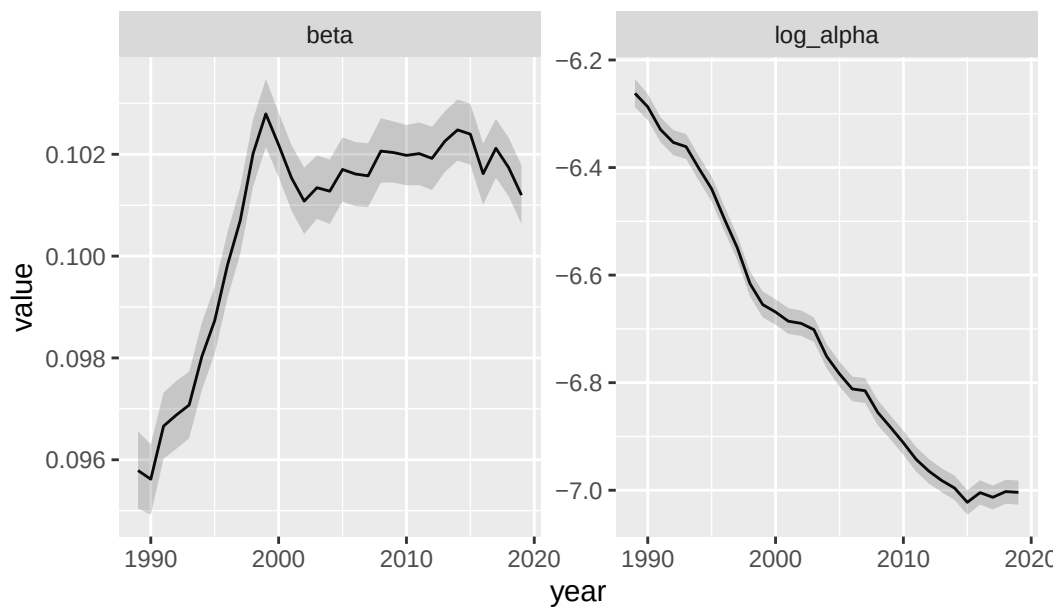



5.1 Plot parameter estimates over time

Now we can plot the parameter estimates over time:

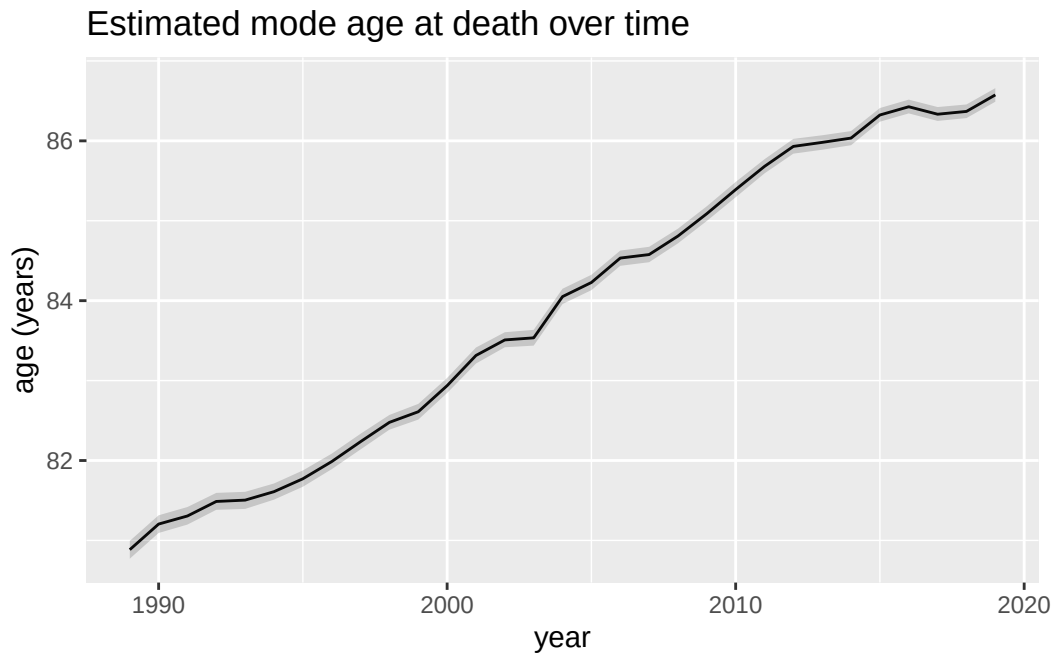
```
mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, .value)) + geom_line() +
  facet_wrap(~.variable, scales = "free_y") +
  geom_ribbon(aes(ymin = .lower, ymax = .upper), alpha = 0.2)+
  labs(y = "value", title = "Estimates of Gompertz parameters over time")
```

Estimates of Gompertz parameters over time



...and also the mode age over time:

```
mod |>
  spread_draws(log_alpha[i], beta[i]) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))+40) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, mode_age)) + geom_line() +
  geom_ribbon(aes(ymin = mode_age.lower, ymax = mode_age.upper), alpha = 0.2) +
  labs(title = "Estimated mode age at death over time", y = "age (years)")
```



5.2 Question

Can you forecast mortality rates with this model? If so, how? What are the assumptions behind the forecasts?

6 Partial data observed

Now we're going to switch gears a bit and have a look at the situation where we have mortality rates for five geographic areas, but in one area we only have partial information. This is based on simulated data (you can have a look to see how I generated it based on the `simulated_data.R` script).

For 4 areas, we have deaths and population counts from ages 40 up to 60:

```
df <- read_rds("../data/sim.rds")
df
```

```
# A tibble: 80 x 4
# Groups:   area [4]
   age area deaths pop
<int> <int> <int> <dbl>
1    40     1     1 5000
```

```

2    41    1    1  5000
3    42    1    1  5000
4    43    1    2  5000
5    44    1    1  5000
6    45    1    2  5000
7    46    1    0  5000
8    47    1    3  5000
9    48    1    5  5000
10   49    1    3  5000
# i 70 more rows

```

For one region, we just have ${}_{20}q_{40}$, that is, the probability of dying between ages 40 and 60:

```

q40 <- read_rds("../data/q40.rds")
q40

```

```
[1] 0.07262754
```

6.1 Relating ${}_{20}q_{40}$ to the Gompertz model

We want to fit a Gompertz model to each of the five areas (even the one with just a summary indicator). How to do this? Well, if we tell Stan how ${}_{20}q_{40}$ relates to μ_x , then the model has at least some information to estimate α and β . In particular, for each area, we're assuming:

$${}_1p_x = e^{-\mu_x}$$

where $\mu_x = \alpha e^{\beta x}$ and

$${}_{20}q_{40} = 1 - \prod_{x=40}^{60} {}_1p_x$$

Check the Stan file `gomp_partial.stan` to see this translated into code.

6.2 Fit the model

Get the data in the right format. Note that everything is a matrix now because we have more than one area:

```

y <- as.matrix(df |>
  pivot_wider(names_from = "area", values_from = "deaths") |>
  select(-age, -pop))
pop <- as.matrix(df |>
  select(-deaths) |>
  pivot_wider(names_from = "area", values_from = "pop") |>
  select(-age))

stan_data <- list(N = length(40:59),
  M = 4,
  K = 5,
  y = y,
  pop = pop,
  z = q40,
  age_c = (40:59)-40)

```

Fit the model:

```

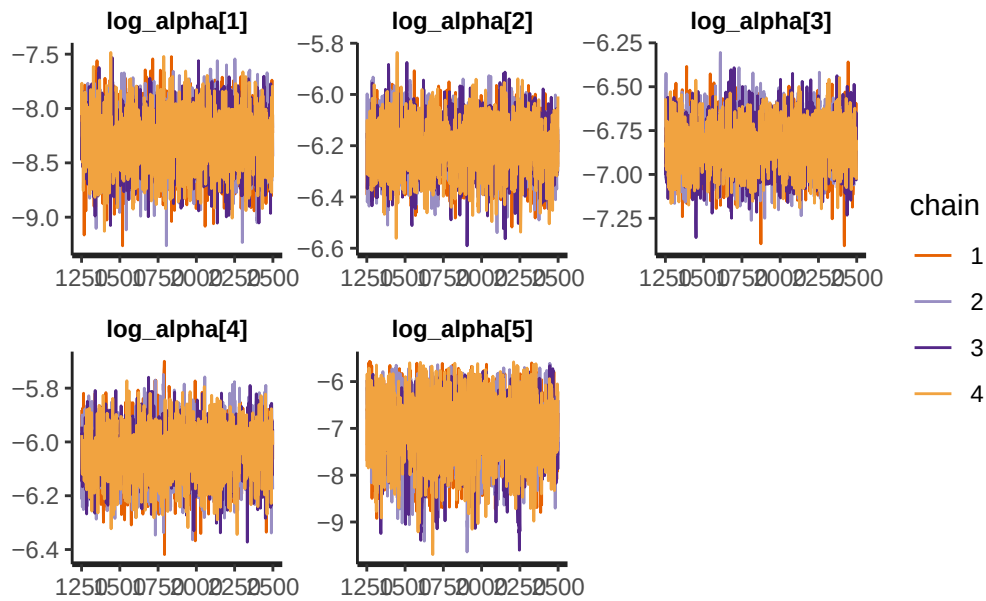
mod <- stan(file = "../code/models/gomp_partial_data.stan",
  data = stan_data,
  seed = 15,
  control = list(adapt_delta = 0.96),
  iter = 2500,
  refresh = 0)

```

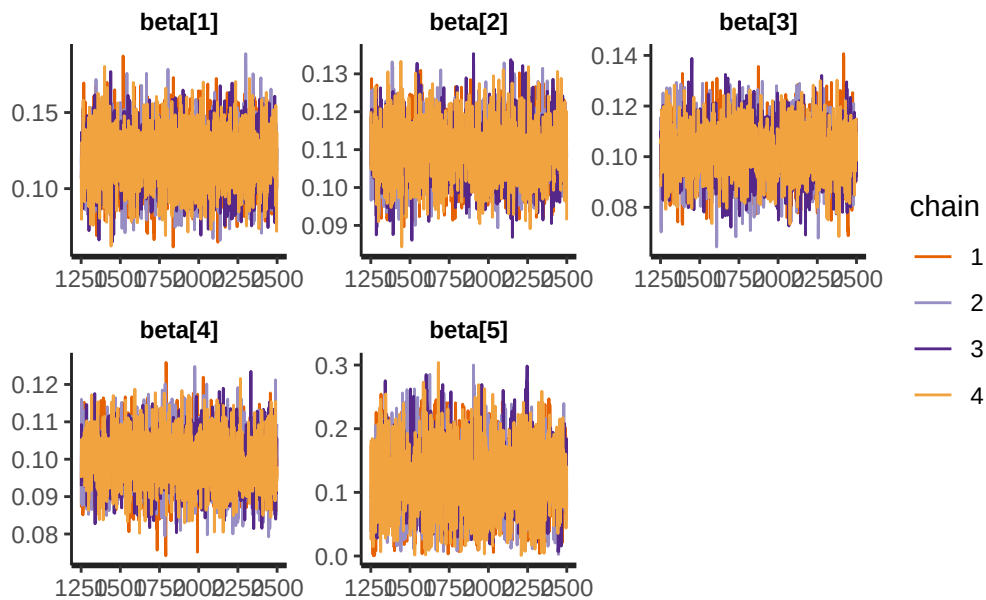
```

traceplot(mod, c("log_alpha"))

```



```
traceplot(mod, c("beta"))
```



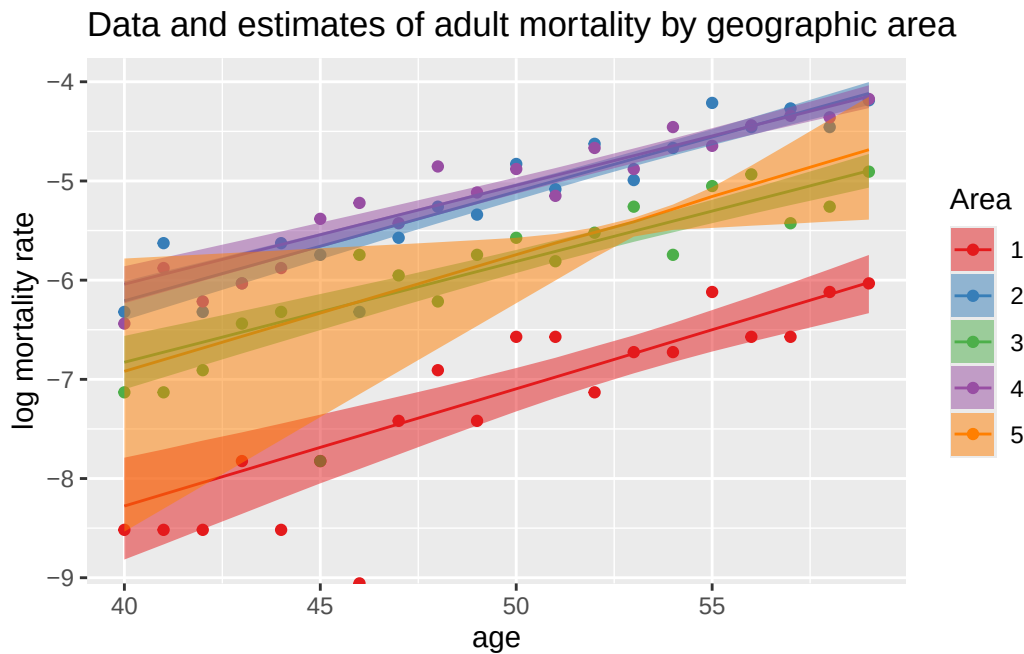
Let's plot the fitted lines (with uncertainty) with the observed data for each region. Notice the difference in uncertainty around area 5!

```
mod |>
  gather_draws(mu[i,j]) |>
  median_qi() |>
```

```

mutate(age = i-1+40) |>
rename(area = j) |>
left_join(df) |>
mutate(log_mx = log(deaths/pop)) |>
ggplot(aes(age, log_mx))+
  geom_point(aes(color = factor(area)))+
  geom_line(aes(age, .value, color = factor(area)))+
  geom_ribbon(aes(age, ymin = .lower, ymax = .upper, fill = factor(area)), alpha = 0.5)+
  scale_color_brewer(name = "Area", palette = "Set1")+
  scale_fill_brewer(name = "Area", palette = "Set1")+
  labs(title = "Data and estimates of adult mortality by geographic area", y = "log mortality")

```



6.3 Compare parameter estimates against the truth

Here's what we estimated:

```

mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi()

```

```

# A tibble: 10 x 8
  i .variable .value .lower .upper .width .point .interval

```

	<int>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	1	beta	0.118	0.0831	0.156	0.95	median	qi
2	1	log_alpha	-8.28	-8.82	-7.79	0.95	median	qi
3	2	beta	0.110	0.0957	0.125	0.95	median	qi
4	2	log_alpha	-6.21	-6.40	-6.01	0.95	median	qi
5	3	beta	0.102	0.0815	0.121	0.95	median	qi
6	3	log_alpha	-6.83	-7.10	-6.56	0.95	median	qi
7	4	beta	0.0996	0.0860	0.113	0.95	median	qi
8	4	log_alpha	-6.04	-6.23	-5.86	0.95	median	qi
9	5	beta	0.118	0.0205	0.231	0.95	median	qi
10	5	log_alpha	-6.92	-8.54	-5.78	0.95	median	qi

And here's the truth (the values underlying the simulation):

```
read_rds("../data/true_params.rds")
```

```
# A tibble: 5 x 3
  are log_alpha beta
<int>   <dbl> <dbl>
1     1    -7.98 0.0985
2     2    -6.38 0.125
3     3    -6.88 0.110
4     4    -5.93 0.0924
5     5    -7.48 0.154
```